

Problem Statement ID : SIH26018

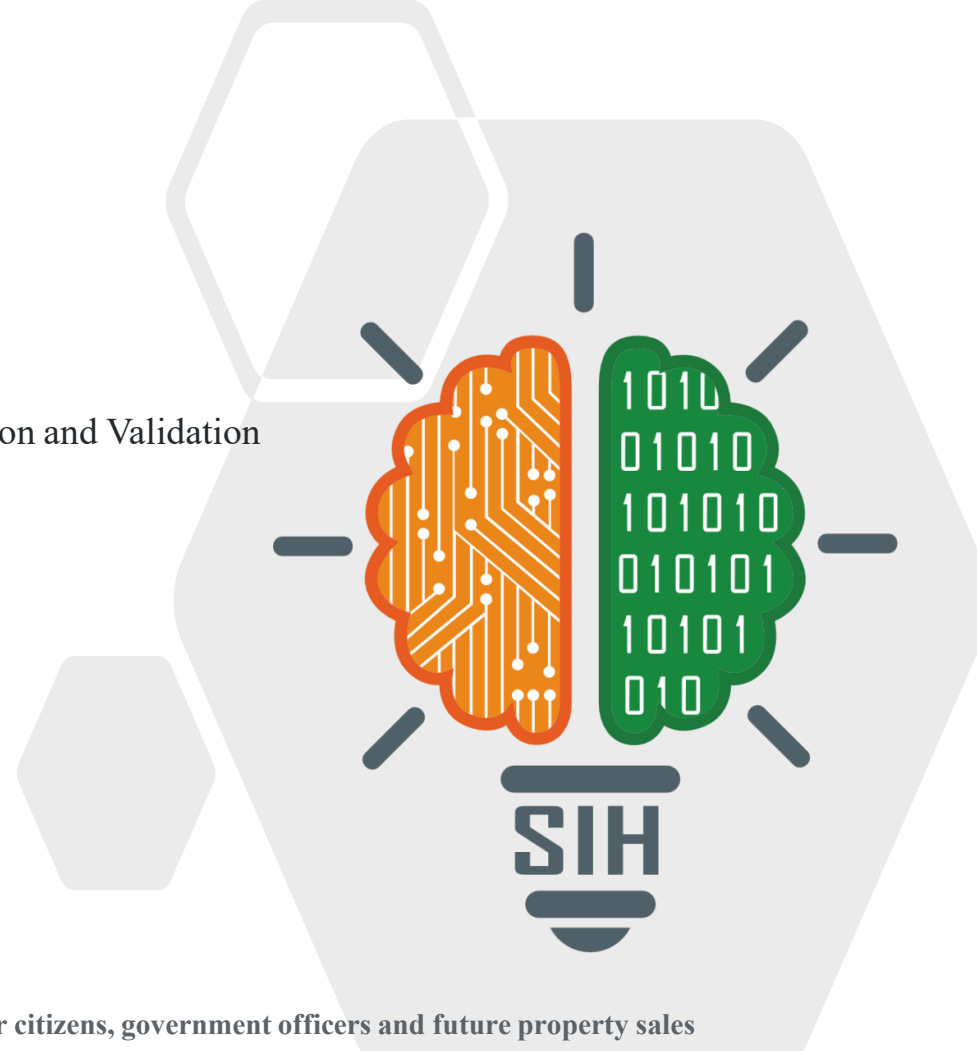
Problem Statement Title : Intelligent Land Record Digitization and Validation System

Theme : Smart Automation

PS Category : Software

Team ID : SIH26-A0H-T309

Team Name (Registered on portal) : **SIXTH SENSE**



One point evidence driven layer for citizens, government officers and future property sales

EXTRACT → VALIDATE → SUMMARIZE → EXPLAIN → PRIORITISE → ACT

PROBLEM AND PROPOSED SOLUTION

Problem • prototype • differentiation • key capabilities

PROBLEM AT HAND

- FRAGMENTED EVIDENCE**
Deeds, RoR, mutation, registration, maps and legacy files are currently existing in different portal
- HUMAN INTERPRETATION GAP**
A citizen may possess documents but still cannot tell whether they agree or what a mismatch means importantly, since written in old, complex language.
- MANUAL OFFICER WORK**
Staff must compare fields, dates and spatial evidence repeatedly, manually, before deciding which cases deserve attention, which consumes a lot of time
- MARKET INFORMATION ASYMMETRY**
Buyers depend on brokers, local claims or advertisements because property evidence is difficult to understand and compare.

OUR SOLUTION — WEB PROTOTYPE

PROPERTY INTELLIGENCE

92% READY

2.50 acres • Survey 142/3 • Coimbatore

	RoR	SALE DEED	EC	MAP
FIELD				
Owner		Ravi Kumar	MATCH	Identity aligned
Survey		142/3	MATCH	Parcel ID aligned
Area		2.50 vs 2.31 ac	REVIEW	7.6% difference
Timeline		2019→2023	MATCH	No break detected

AI EXPLAINS TO THE USER:
 “Survey area is 7.6% lower. This may indicate a measurement, subdivision/update or boundary-record discrepancy. Verify the latest survey evidence.”

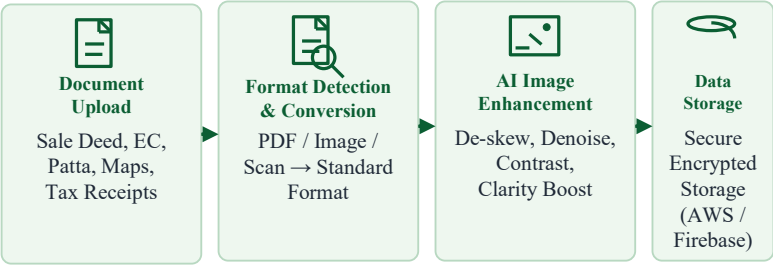
WHY WE STAND OUT and KEY FEATURES

- Property Intelligence Profile**
One structured profile links owner, parcel, documents, events and spatial evidence.
- Evidence-linked AI**
Every important finding points back to the fields/documents that produced it.
- Conflict reasoning**
Our project not just shows ‘mismatch’ in uploaded data but also specifies where the actual discrepancy is.
- Citizen + Officer modes**
Same evidence engine provides a simple explanation for the public also acts as an investigation queue for officers.
- Marketplace-ready trust**
Verified details and documents become the basis for buyer–property matching, rather than relying only on price/ads.

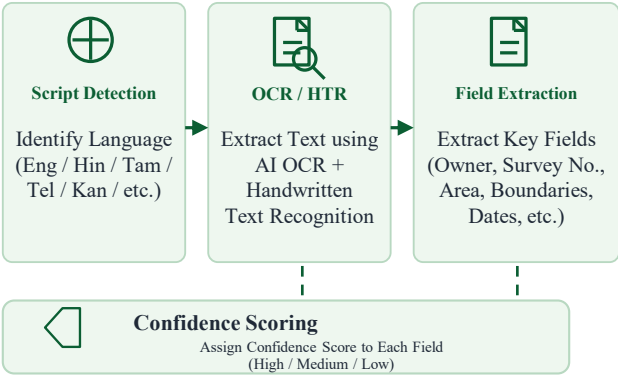
MARKETPLACE — SAME AI ENGINE, NEW DECISION



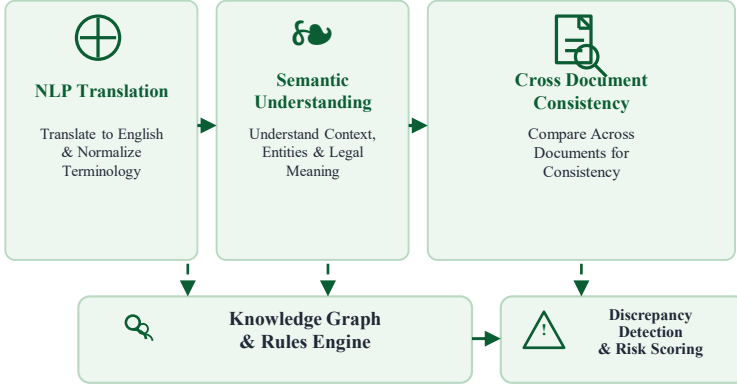
1. INPUT PROCESSING



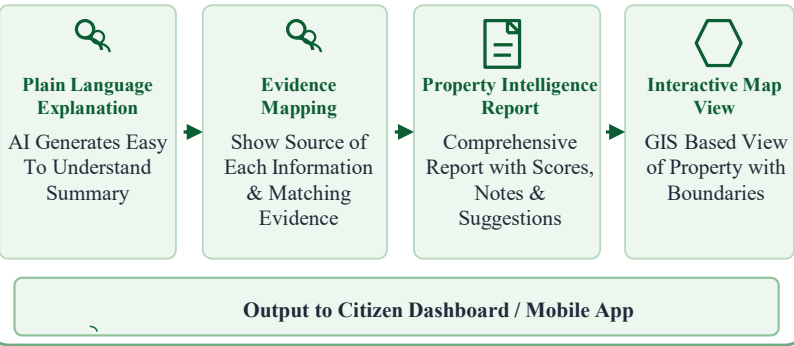
2. TEXT DETECTION & OCR



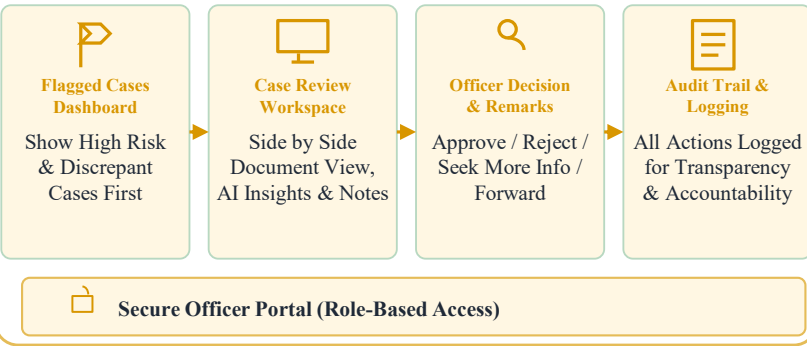
3. UNDERSTANDING & VALIDATION



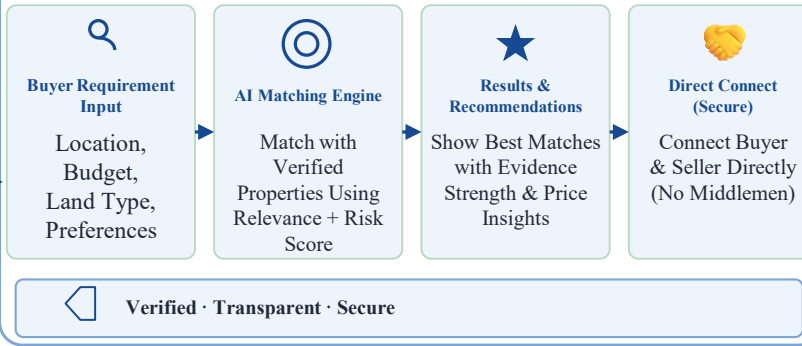
4. AI EXPLANATION & CITIZEN OUTPUT



5. GOVERNMENT OFFICER WORKFLOW



6. VERIFIED MARKETPLACE MATCHING



TECH STACK

FRONTEND

Flutter
Dart

BACKEND

FastAPI
Python

AI / ML

PyTorch
Transformers (Hugging Face)
OpenCV

NLP

mBERT / IndicBERT
spaCy

DOCUMENT AI

Tesseract OCR
EasyOCR

DATABASE

MongoDB
Firebase

CLOUD & STORAGE

AWS
Firebase Storage

MAPS & GIS

Mapbox
GeoPandas

OTHER TOOLS

Docker
GitHub Actions
Postman

FEASIBILITY AND VIABILITY

Technical feasibility • implementation • safety • before / after

TECHNICAL FEASIBILITY

- ✓ OCR / Computer Vision / NLP / GIS are commercially available building blocks.
- ✓ Deterministic rules handle measurable checks; AI handles classification and explanation.
- ✓ Small labelled prototype can demonstrate the complete pipeline.
- ✓ Spatial comparison can begin with sample datasets of surveyed plots

DATA ACCESS FEASIBILITY

- ✓ PROTOTYPE → 100% Uses permitted imaginary datasets of a person
- ✓ PILOT → authorised departmental datasets / APIs.
- ✓ PRODUCTION → connect to existing government LRMS, DILRMP, GIS, registration and other authorised databases through approved APIs/data-sharing.
- ✓ Our platform is an intelligence layer on top of official records already available, not a replacement database.

ECONOMIC or OPERATIONAL VIABILITY

FACTOR	PROTOTYPE	SCALE-UP
Compute	Local / cloud	GPU / cloud
Data	Sample corpus	Authorised feeds
Review	Human queue	District workflow
Storage	Object store	Gov-grade infra
Revenue	Pilot / grant	SaaS / API / services

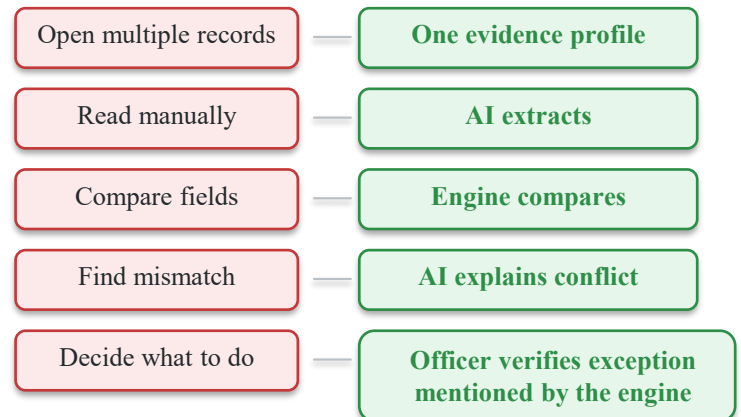
IMPLEMENTATION VIABILITY

- ✓ Our project is just a simple one point intelligent web/app version that can be easily accessible to the general public or by government officials.
- ✓ The project can be expanded in a wider national scale as the dataset increases

MARKET COMPETITION

- Companies like Magicbricks, Jaaga, Deedwise are already trying to find better solution for this problem.
- Their existing solution is widely useful for big land contractors and companies and is not trustable source for general public since they focus results based on ads based model rather than actual verified documents

BEFORE → AFTER



IMPACT AND MARKET VALUE

Impact • India scale • government value • marketplace

INDIA SCALE

6.30 LAKH

villages / CLR

4.02 CRORE

digital maps

4,148

SROs computerised

DILRMP 3.0

2026–2031

Official DoLR dashboard / programme

TARGET AUDIENCE

COMMON PEOPLE

• India has an enormous land-record ecosystem: the government reports around **40.1 crore Records of Rights (RoRs)** and nearly **48.5 crore recorded landowners** in its current national land-record statistics. Sixth Sense can act as an intelligence layer that converts scattered records into a simpler, understandable property profile for ordinary citizen

GOVERNMENT- OFFICERS

• Land records are already being digitised at a massive scale—DILRMP reports land-record computerisation at approximately **99.9%**, but connecting textual records with spatial cadastral maps remains a separate challenge. Our system focuses on intelligently comparing available records and explaining inconsistencies, potentially supporting officers during document and case review.

BUSINESS ASPECT

• The Government's own vision for DILRMP includes an integrated land-information system that benefits both **landowners and prospective buyers**, reduces physical visits, helps prevent fraudulent transactions and enables information sharing. Sixth Sense extends this concept into a **trust-oriented marketplace**, where property discovery can be connected with documentary intelligence rather than relying only on advertisements.

MARKET VALUE

• **India's PropTech market was estimated at approximately USD 1.66–1.72 billion in 2025** (roughly ₹14,000–15,000 crore) and is projected by different market-research estimates to grow strongly over the coming years.

• **Sixth Sense** targets this India's rapidly growing PropTech market, valued at approximately USD 1.7 billion (around ₹14,000 crore)

- **Government Land Records & Digital Infrastructure**

- [Department of Land Resources \(DoLR\), Government of India](#)
- [Digital India Land Records Modernization Programme \(DILRMP\)](#)
- [National Generic Document Registration System \(NGDRS\)](#)
- [Digital India – Government Digital Infrastructure](#)

- **Land Mapping, Survey & Geospatial Research**

- [ISRO Bhuvan – Indian Geo-Platform](#)
- [Survey of India – National Mapping Agency](#)
- [GeoPandas Documentation – Geospatial Data Analysis](#)
- [Mapbox Documentation – Interactive Maps and Location Intelligence](#)

- **Document AI, OCR & Intelligent Data Extraction**

- [Tesseract OCR – Open-Source OCR Engine](#)
- [EasyOCR – Deep Learning-Based OCR Framework](#)
- [PaddleOCR – Intelligent Document Processing](#)
- [Hugging Face Transformers Documentation](#)

- **AI Validation, Explainability & Property Intelligence**

- [PyTorch – Machine Learning Framework](#)
- [IBM AI Explainability 360 – Explainable AI Toolkit](#)
- [JaaGa – Property Intelligence Platform](#)

- **Property Marketplace & Industry Research**

- [Magicbricks – Digital Property Marketplace](#)
- [NoBroker – Direct Property Transaction Platform](#)
- [Housing.com – Digital Real Estate Platform](#)

- **Technical Development References**

- [FastAPI Official Documentation](#)
- [MongoDB Official Documentation](#)
- [Firebase Official Documentation](#)

Thank You

SCAN TO GET LIVE EXPERIENCE